Reg. No. | | | | | | | | | | | [13EC62/12EC62]

SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE – 2016

INFORMATION THEORY AND CODING

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any FIVE full questions

1. a. Define Entropy. List any five properties of Entropy with relevant equations. **06**

   b. For the I order Markov Source shown in figure 1(b), find **08**

   i. State Probabilities

   ii. Entropy of each State

   iii. Entropy of the Source

   iv. G_1 and G_2 and verify that $G_1 > G_2 > H(S)$

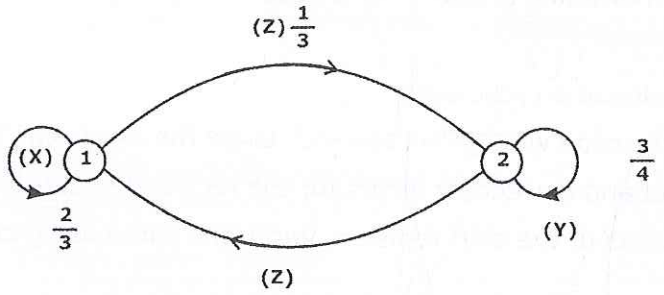

Figure 1(b)

2. a. A discrete memory less source with 8 alphabets A to H has respective probabilities 0.22, 0.2, 0.18, 0.15, 0.10, 0.08, 0.05 and 0.02. Construct a binary compact Huffman Code and determine the code efficiency. **08**

   b. Find the smallest number of letters in the alphabet for devising a code with prefix property such that {w} = {0, 3, 0, 5}, where w is the set of number of words with word length 1, 2, 3,.... **04**

   c. Given, code A= {0, 100, 110, 11}, find whether code A is instantaneous. **02**

3. a. For the JPM given below, compute individually H(X), H(Y), H(X,Y), H(Y/X), H(X/Y) and I(X,Y). **08**

$$p(x,y) = \begin{bmatrix} 0.05 & 0 & 0.20 & 0.05 \\ 0 & 0.10 & 0.10 & 0 \\ 0 & 0 & 0.20 & 0.10 \\ 0.05 & 0.05 & 0 & 0.10 \end{bmatrix}$$

   b. What is a Binary erasure channel? Obtain an expression for its channel capacity. **06**

4. a. Show that $I(A,B) \geq 0$. **04**

b. State and explain Shannon-Hartley law. Derive the expression for the upper limit of the channel capacity. **06**

c. Consider an AWGN channel with 4kHz bandwidth and noise power spectral density $n/2 = 10^{-12}$watts/Hz. The signal power required at the receiver is 0.1mW. Calculate the capacity of the channel. **04**

5. a. Design a single error correcting code with a message block size of 11 and show that by an example that, it can correct a single error. Find [G] and [H] of this code. **08**

b. If 'C' is a valid code-vector, then prove that, $C.H^T=0$, where H^T is the transpose of the parity check matrix-H. **04**

c. Explain the term "Hamming distance" of a code. **02**

6. a. List all the properties of a cyclic code. **04**

b. Given a (7,4) cyclic code with $g^{(x)}=1+x=+x^3$. Draw the syndrome calculation circuit. Also detect and correct for errors for the received vector R=1101001 by showing the contents of the shift register. Verify the same using conventional method. **10**

7. a. Draw a block diagram of a (2, 1, 3) conventional encoder with impulse responses $g^{(1)}=1011$ and $g^{(2)}=1101$. Find the code using both time domain and transform domain approaches for the input data D=11011. **10**

b. What is a code tree? Explain with an example. **04**

8. a. Write short notes on: **08**

i. R-S Codes

ii. Golay Codes

b. What is a standard array? Explain with an example. **06**